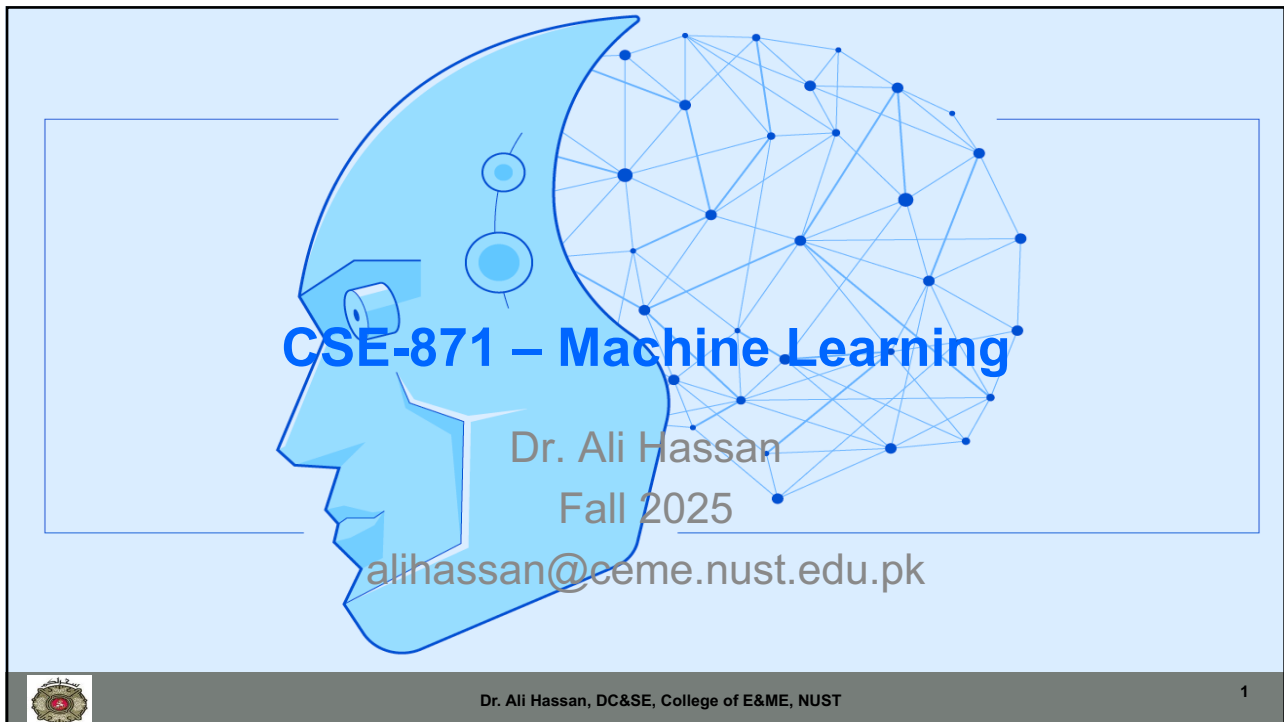


A graphic for the CSE-871 – Machine Learning course. It features a blue profile of a human head facing right. Inside the head, there are two circular nodes connected by lines, and a larger, more complex network of nodes and lines representing a neural network. The text "CSE-871 – Machine Learning" is written in blue across the middle of the head. Below it, in brown, is "Dr. Ali Hassan", "Fall 2025", and "alihassan@ceme.nust.edu.pk". At the bottom left is a small NUST logo. At the bottom right is the number "1".

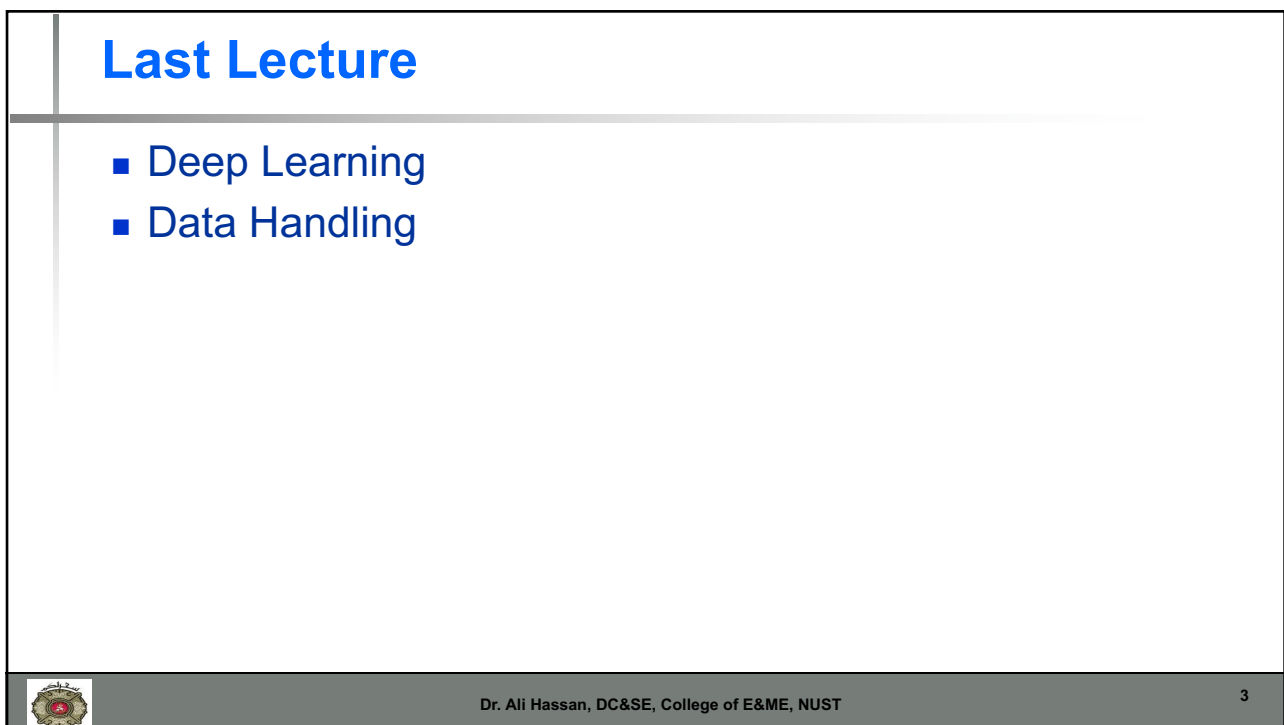
**CSE-871 – Machine Learning**

Dr. Ali Hassan  
Fall 2025  
alihassan@ceme.nust.edu.pk

Dr. Ali Hassan, DC&SE, College of E&ME, NUST

1

1

A slide titled "Last Lecture" in blue. Below the title, there is a list of two items: "■ Deep Learning" and "■ Data Handling". At the bottom left is a small NUST logo. At the bottom right is the number "3".

## Last Lecture

- Deep Learning
- Data Handling

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3

3

## Outline

- Autoencoders
- Regularised Autoencoders
- Denoising Auto Encoders
- Variational AutoEncoders -- GAN

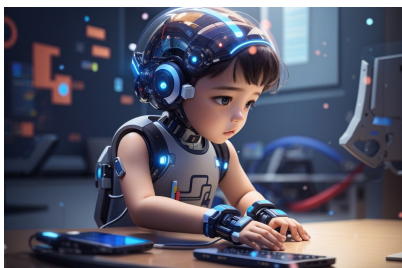


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## Future of Kids with Artificial Intelligence



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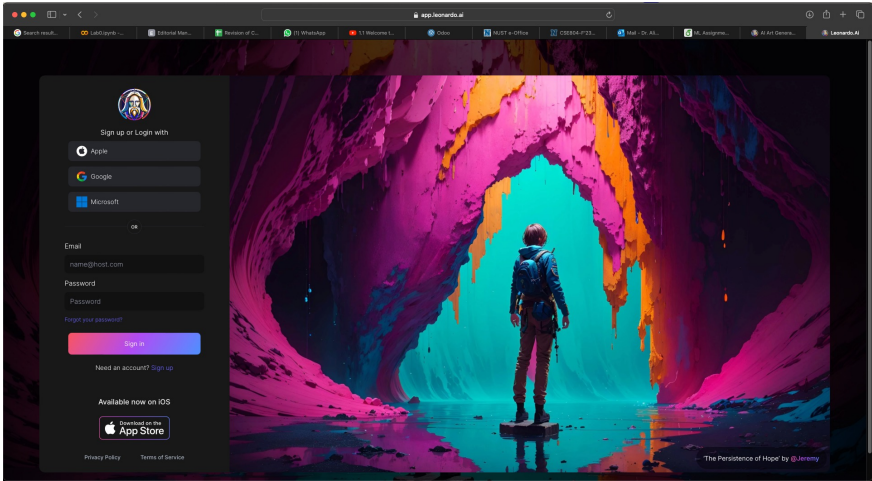
# Draw Faisal mosque



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# Leonardo.ai



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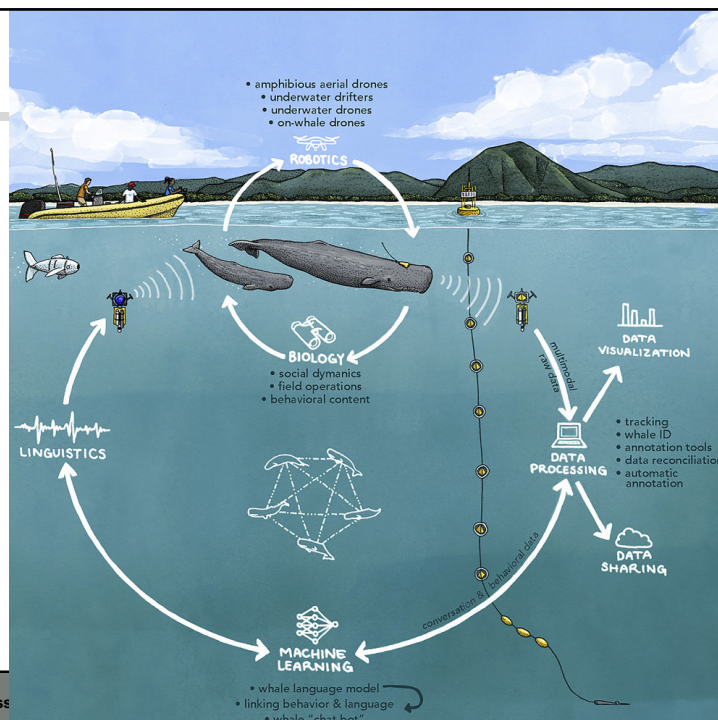
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## Data Acquisition

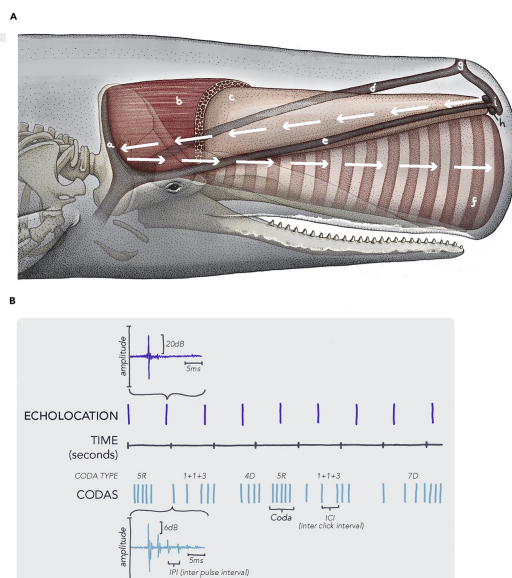


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## Echolocations and Coda Clicks

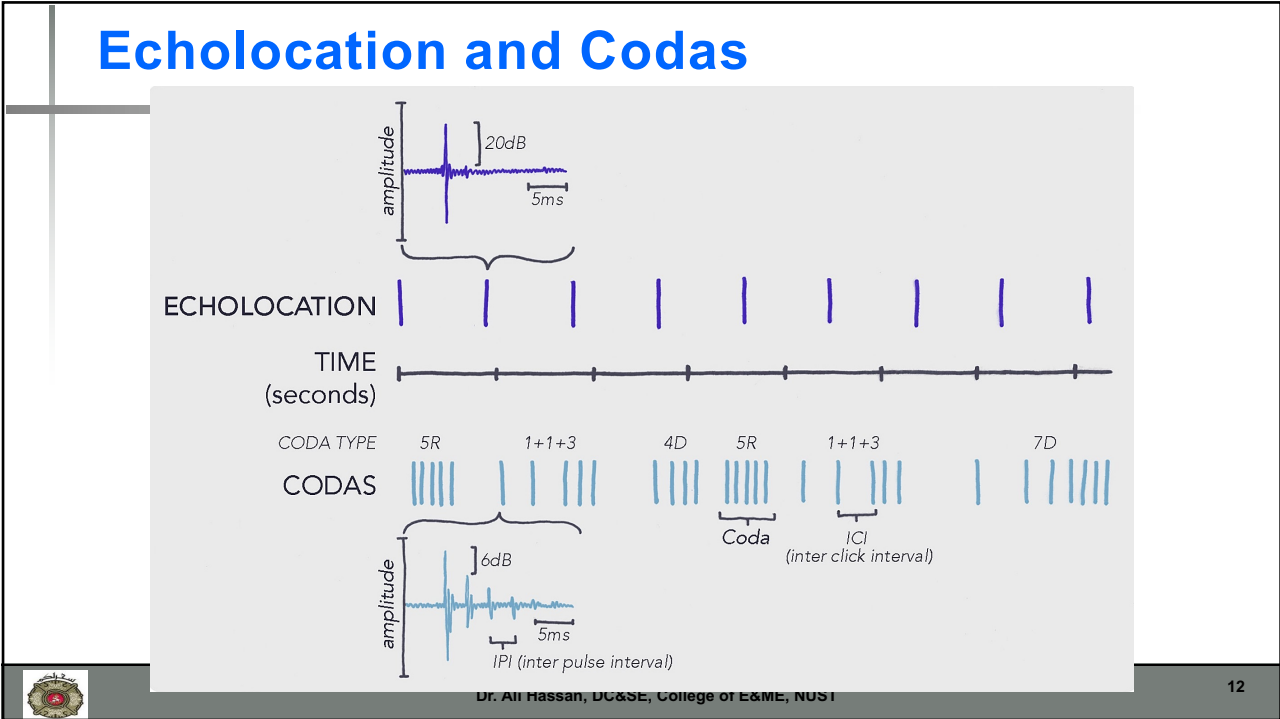
- Sound consists of echolocation and coda clicks
- Echolocation produced with inter-click intervals (0.4 s)
- Coda clicks are arranged in sequences called "codas"
- Codas are characterized by the different number of constituent clicks and the intervals between them (called inter-click intervals or ICIs).
- Codas are typically produced in multi-party exchanges
- Last from about 10 s to over half an hour.
- Each click, is a sequence of equally spaced pulses, with inter-pulse interval (IPI) of an order of 3–4 ms in an adult female



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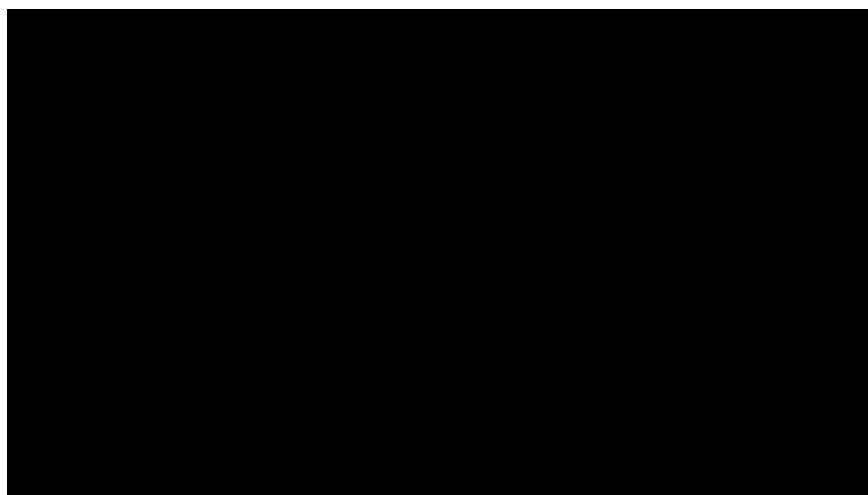


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<https://youtu.be/a8F53P3VA1c?si=xLMC--wSWzb-s-eb>



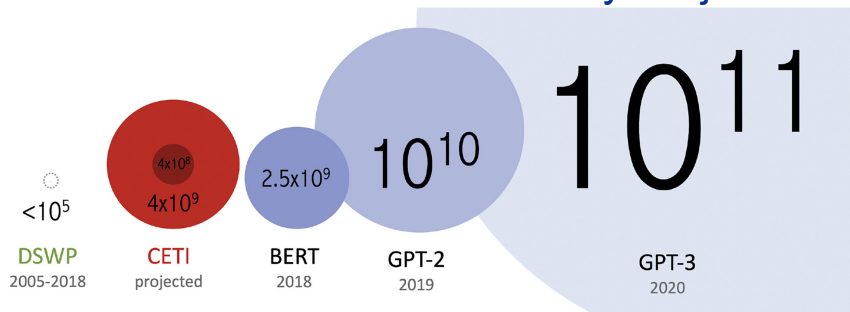
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## Large Language Model for Dominica Project

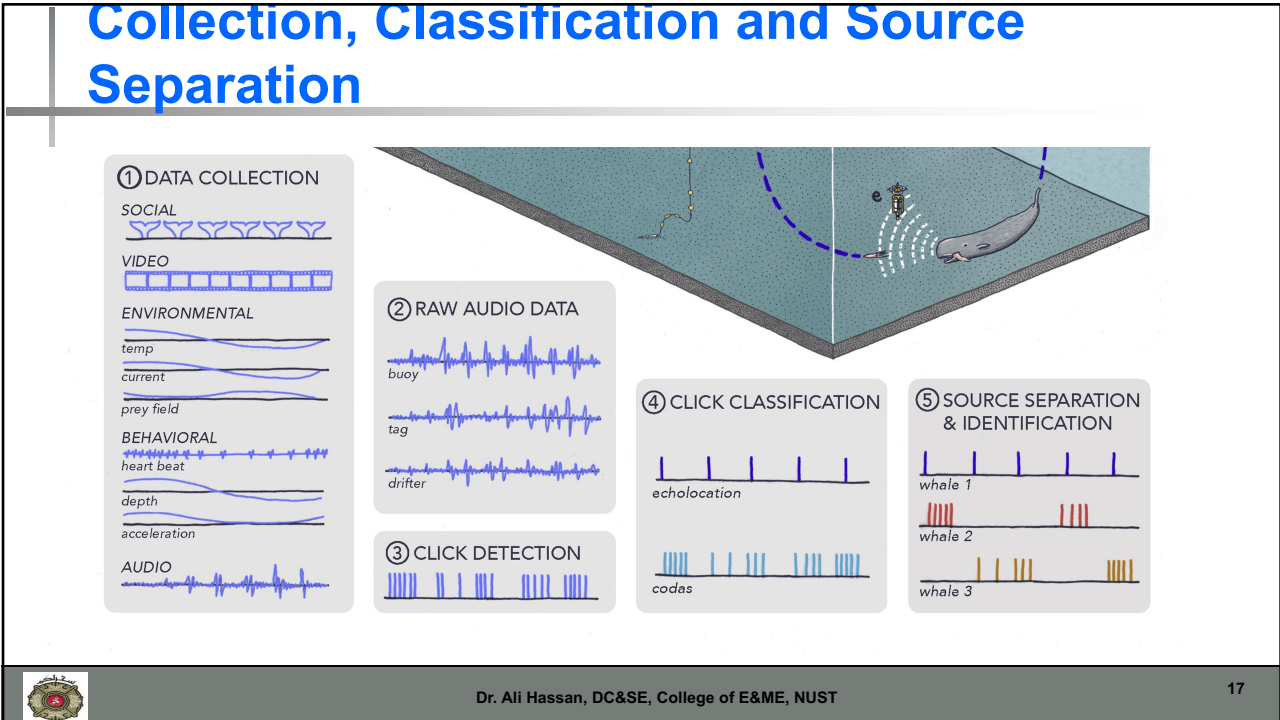
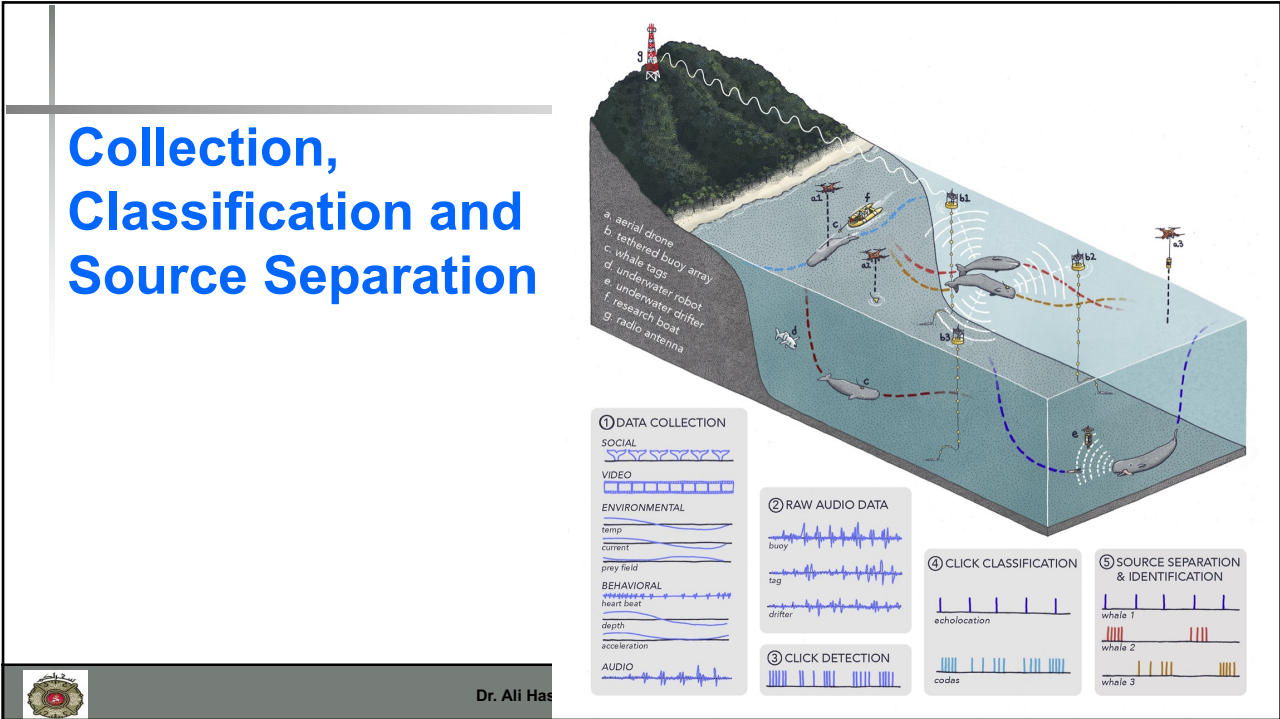
- while the DSWP dataset is a tiny dot on this plot (located at the center of the dashed circle).
- Shown in red is the estimated size of a new dataset planned to be collected in Dominica by Project CETI



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HOME / SPERM WHALES: A BABY IS BORN

## Sperm Whales: A Baby Is

TRAVEL & TRIPS ECOLOGY & SCIENCE — Faial Island, Azores, Septem cruised the islands of Faial and on the small boat is tiresome and long, but necessary for cl sperm whales of the Atlantic.

Kurt Amisler



support her newborn, which is easily 2.6 metres in length.

With every passing minute the baby is increasingly mobile and able to swim independently over short distances. I can also hear his communication, which has a higher pitch than the others—the voice of a child.

### Protective mother

In order not to disturb the animals, I move very carefully and maintain a distance of about 10 metres. Up to this point, the whales have not taken any notice of my presence, but now it seems the mother wants to identify the stranger in their midst.

Quietly but directly, the nine-metre long giant turns in my direction and swims right up to me. Her massive head is getting bigger and bigger as the water displacement pushes me away. Water churns around me and the exhalation noise thunders in my ears! I see the eye looking at me and I feel absolutely no aggression. I am thoroughly overwhelmed by the experience!

Whales communicate perpetually, with their sounds being heard by others over great distances. As the birth was communicated, more and more animals arrive for the event. The giants surround me and I am fully accepted!

### Presenting a newborn

The mother swims to other groups to present her child. She does the same to me, stopping and allowing the baby swim towards me—incredible! After about 20 minutes, the baby becomes stronger and moves faster. From time to time, it wants to venture away on its own, which the mother does not like at all. With her immense toothy mouth, she brings the runaway back to surface.

Sperm whales feed on giant squid, which they hunt to depths of 2,000 metres. More than 40 conical teeth, each reaching a length 20cm, sit in the lower jaw.

As the event comes to its conclusion, more and more whales disappear into the blue of the Atlantic. The

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## AI Capturing Brain Activity

### End-to-End Deep Image Reconstruction From Human Brain Activity

Guohua Shen<sup>1†</sup>, Kshitij Dwivedi<sup>1†</sup>, Kei Majima<sup>2</sup>, Tomoyasu Horikawa<sup>2</sup> and Yukiyasu Kamitani<sup>1,2\*</sup>

<sup>1</sup>Computational Neuroscience Laboratories, Advanced Telecommunications Research Institute International, Kyoto, Japan

<sup>2</sup>Graduate School of Informatics, Kyoto University, Kyoto, Japan

Here, we directly trained a DNN model with fMRI data and the corresponding stimulus images to build an end-to-end reconstruction model. We accomplished this by training a generative adversarial network with an additional loss term that was defined in high-level feature space (feature loss) using up to 6,000 training data samples (natural images and fMRI responses). The above model was tested on independent datasets and directly reconstructed image using an fMRI pattern as the input. Reconstructions obtained from our proposed method resembled the test stimuli (natural and artificial images) and reconstruction accuracy increased as a function of training-data size.

<https://www.frontiersin.org/articles/10.3389/fncom.2019.00021/full>



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<https://youtu.be/YrO1v7-KcXs?si=B73b6GI8uHSQYQD4>

RESEARCH ARTICLE

## Deep image reconstruction from human brain activity

Guohua Shen<sup>1</sup>, Tomoyasu Horikawa<sup>1</sup>, Kei Majima<sup>2</sup>, Yukiyasu Kamitani<sup>1,2\*</sup>

<sup>1</sup> Computational Neuroscience Laboratories, Advanced Telecommunications Research Institute International, Kyoto, Japan, <sup>2</sup> Graduate school of Informatics, Kyoto University, Kyoto, Japan

These authors contributed equally to this work.

\* kamitani@i.kyoto-u.ac.jp

### Abstract

The mental contents of perception and imagery are thought to be encoded in hierarchical representations in the brain, but previous attempts to visualize perceptual contents have failed to capitalize on multiple levels of the hierarchy, leaving it challenging to reconstruct



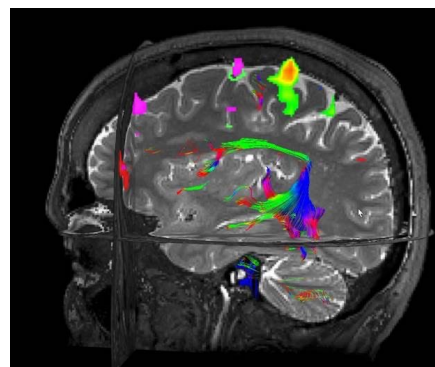
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## fMRI

- Functional magnetic resonance imaging (fMRI) measures the small changes in blood flow that occur with brain activity.



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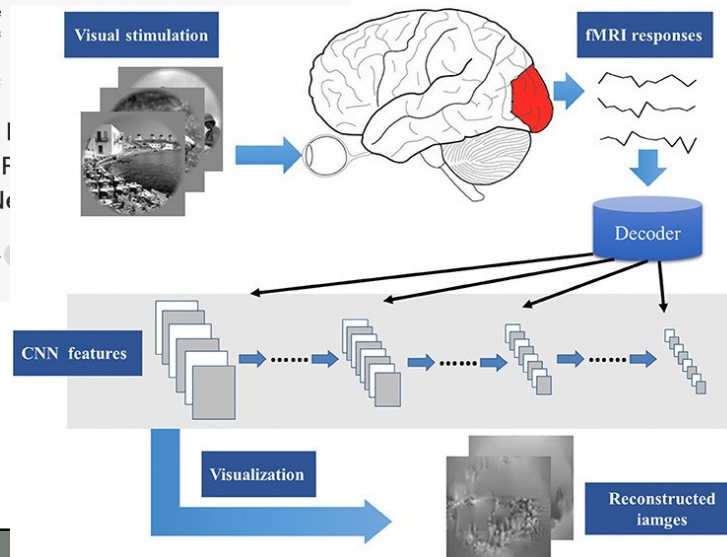
21

## AI Capturing Brain

ORIGINAL RESEARCH article  
Front. Hum. Neurosci., 22 June 2018  
Sec. Sensory Neuroscience  
Volume 12 - 2018 |  
<https://doi.org/10.3389/fnhum.2018.00242>

Constraint-Free  
Reconstruction From  
Convolutional Neural

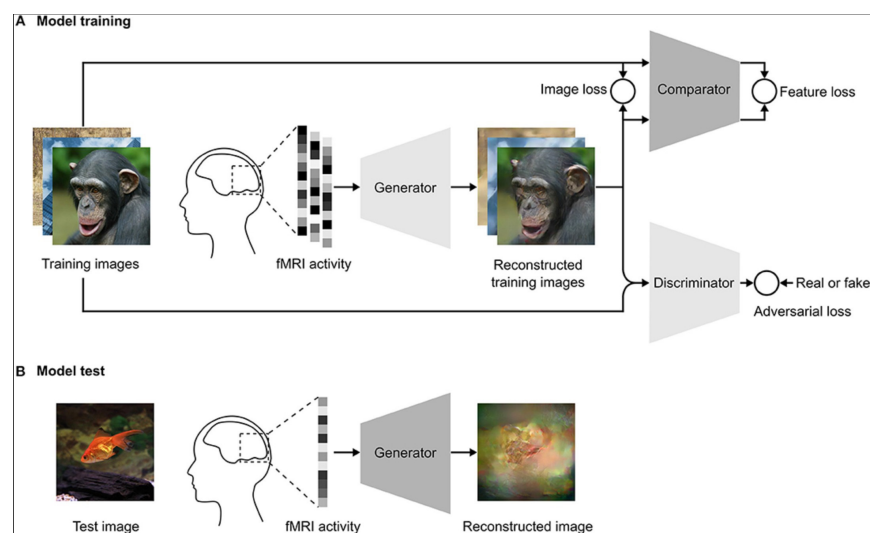
Chi Zhang<sup>1</sup>, Kai Qiao<sup>1</sup>,  
Bin Yan<sup>1\*</sup>



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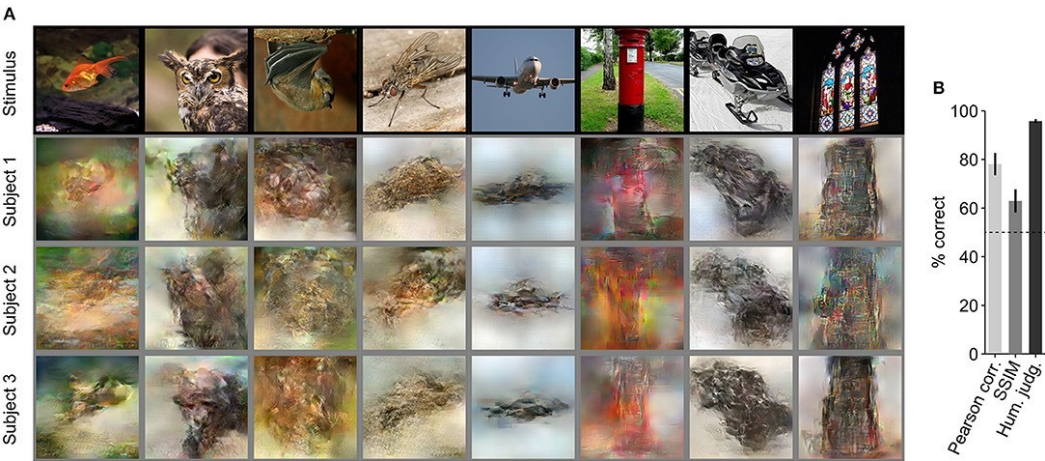
## People will see what you are dreaming using GAN



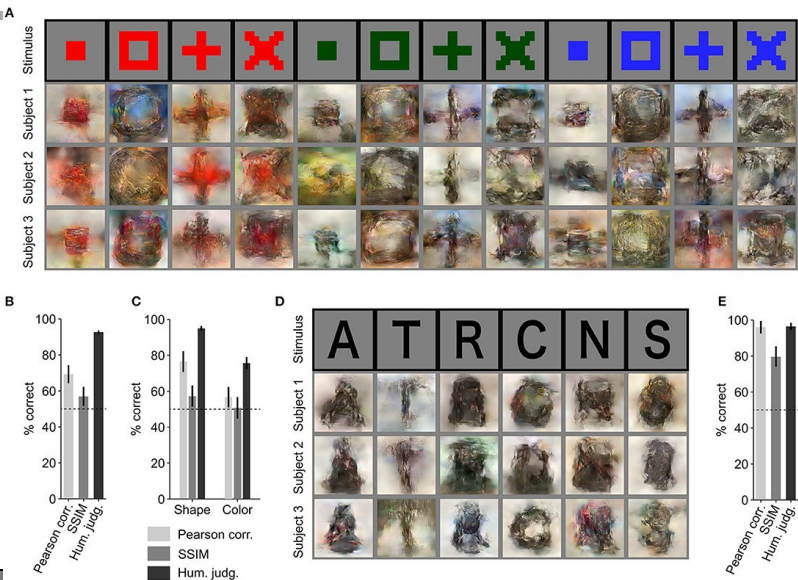
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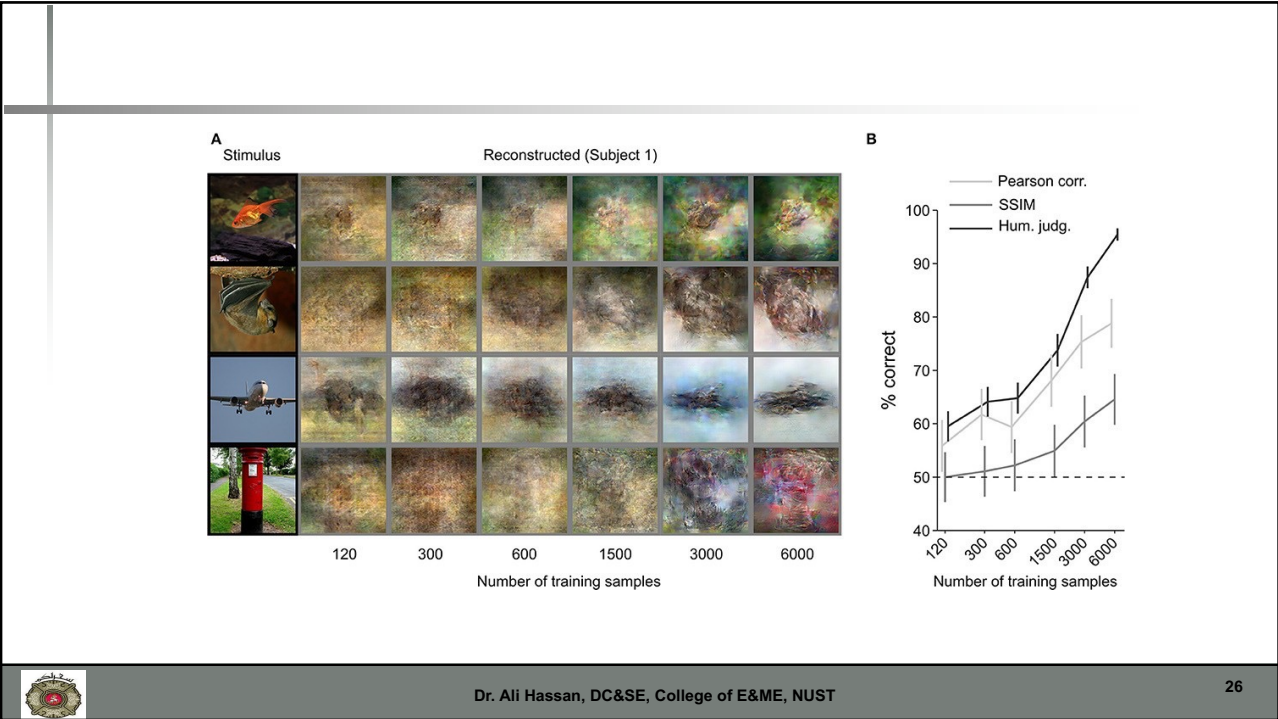
# Gain to Train on Brain Activity



# Gain to Train on Brain Activity







← → ↻ unite.ai/human-image-synthesis-from-reflected-radio-waves/ 🔍 ⌵ ⌵ ⌵ ⌵ ⌵ Relaunch to update

# Human Image Synthesis From Reflected Radio Waves

 Updated on December 9, 2022  
By Martin Anderson



SEARCH

**RECENT POSTS**

- MiniGPT-5: Interleaved Vision-And-Language Generation via Generative Vokens
- How to Protect Your Digital Identity in the Era of AI-Enhanced Imagery
- Revolutionizing Robot Learning: NVIDIA's Eureka Aces Complex Tasks
- 10 Best Password Manager Tools (October 2023)
- Jeff Kofman, Founder & CEO of Trint – Interview Series

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## near photoreal images of people without cameras

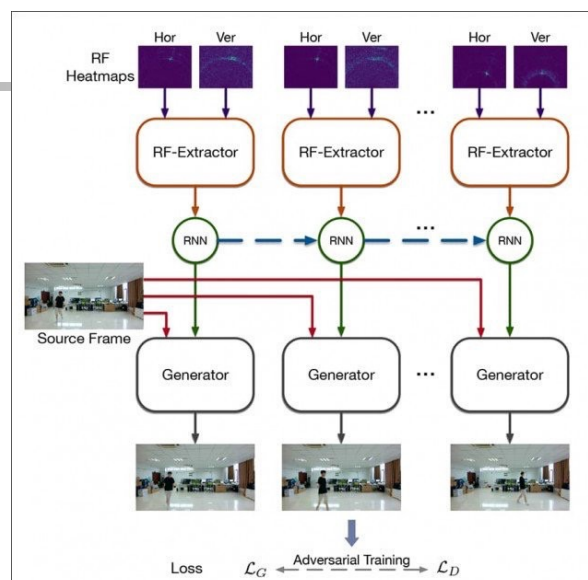
- Researchers from China have developed a method to synthesize near photoreal images of people without cameras
- Using radio waves and Generative Adversarial Networks (GANs)



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### Human poses using Radio signals

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The Washington Post  
*Democracy Dies in Darkness*

Tech Help Desk Artificial Intelligence Internet Culture Space Tech Policy

INNOVATIONS

## They thought loved ones were calling for help. It was an AI scam.

Scammers are using artificial intelligence to sound more like family members in distress. People are falling for it and losing thousands of dollars.

By Pranshu Verma  
March 5, 2023 at 6:00 a.m. EST

Card said in an interview with The Washington Post. “We were convinced that we were talking to Brandon.”

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## Unsupervised Learning

- “We expect unsupervised learning to become far more important in the longer term. Human and animal learning is largely unsupervised: we discover the structure of the world by observing it, not by being told the name of every object.”  
– LeCun, Bengio, Hinton, Nature 2015



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## Unsupervised Learning

- As I've said in previous statements: most of human and animal learning is unsupervised learning. If intelligence was a cake, unsupervised learning would be the cake, supervised learning would be the icing on the cake, and reinforcement learning would be the cherry on the cake. We know how to make the icing and the cherry, but we don't know how to make the cake.  
- Yann LeCun, March 14, 2016 (Facebook)



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## What is an Auto Encoder

- An autoencoder is a neural network that is trained to attempt to copy its input to its output
- It uses supervised learning algorithm, i.e., neural network and backpropagation, to do unsupervised learning.
- It does so by setting the target values to be equal to the inputs. i.e., it uses  $y(i)=x(i)$ .

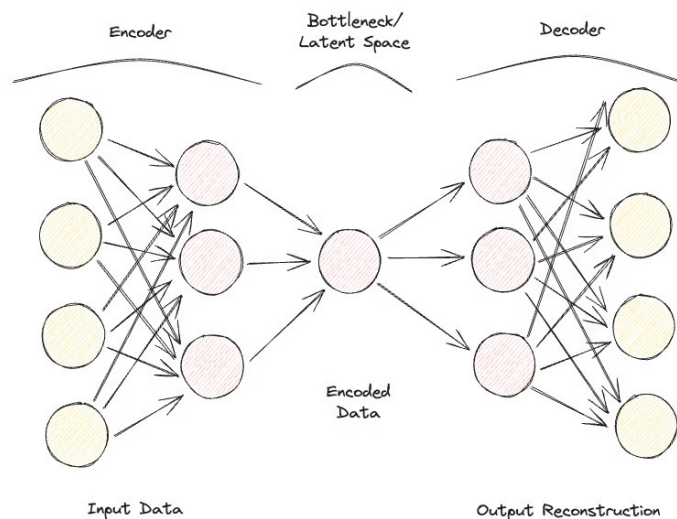


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## Structure of an AutoEncoder



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| Under Represented AE | Over Represented AE |
|----------------------|---------------------|
|                      |                     |



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## Auto Encoders

- The autoencoder tries to learn a function  $h_{W,b}(x) \approx x$ . In other words, it is trying to learn an approximation to the identity function, so as to output  $\hat{x}$  is similar to  $x$ .
- This function seems a particularly trivial function to be trying to learn; but by placing constraints on the network, such as by limiting the number of hidden units, we can discover interesting structure about the data.



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## Applications of AutoEncoders

- Latent Space Representation, Image Compression
- Similar Image Search
  - Reference: Krizhevsky, Alex, and Geoffrey E. Hinton. "Using very deep autoencoders for content-based image retrieval." *ESANN*. 2011.
- GAN
- Segmentation
  - SegNet for Self Driving Cars
- Variational AutoEncoders ...

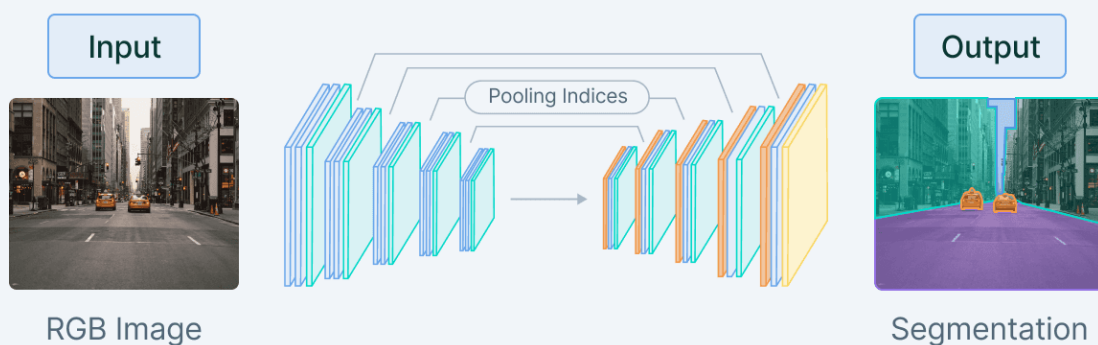


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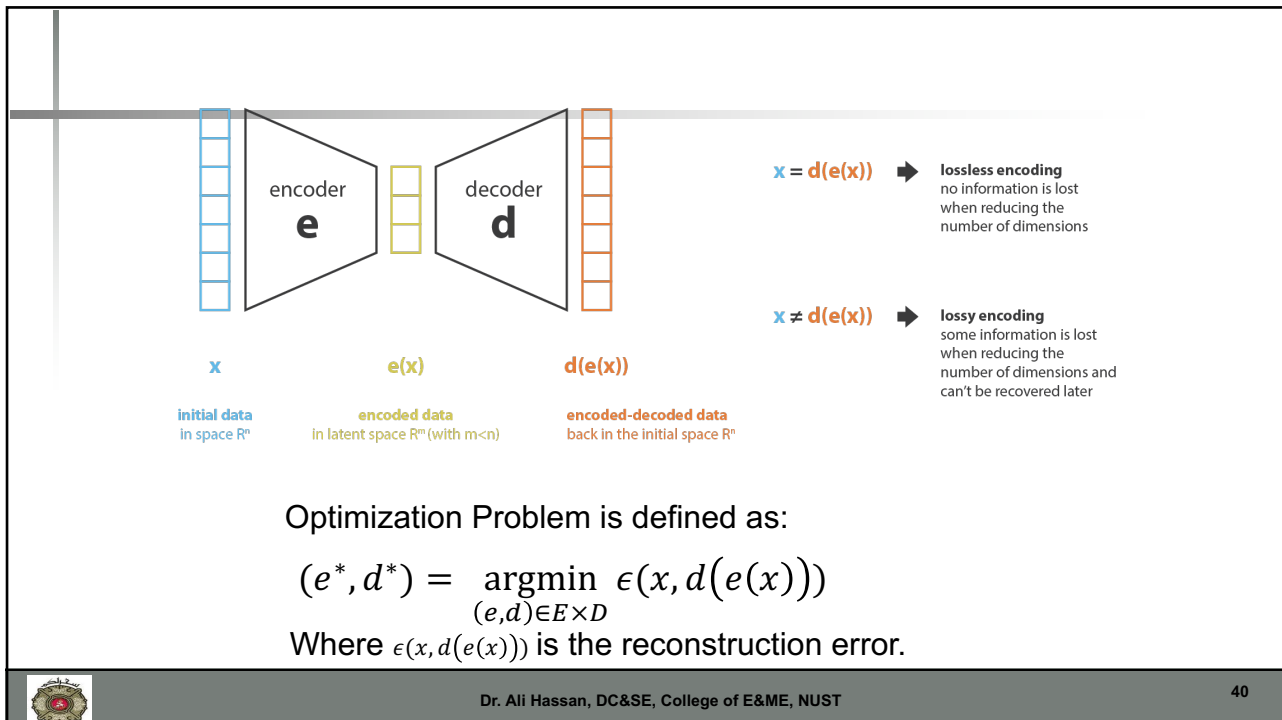
## Convolutional encoder-decoder



V7 Labs

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## Auto Encoder Representation

- Encoder
- Decoder
- Modern autoencoders have generalized the idea of an encoder and a de-coder beyond deterministic functions to stochastic mappings  $p_{\text{encoder}}(h | x)$  and  $p_{\text{decoder}}(x | h)$

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## Similarities with PCA

- $Z = U^T * X \rightarrow$  Encoding
- $X = (U^{-1} * Z) \rightarrow$  Decoding



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## Similarities with PCA

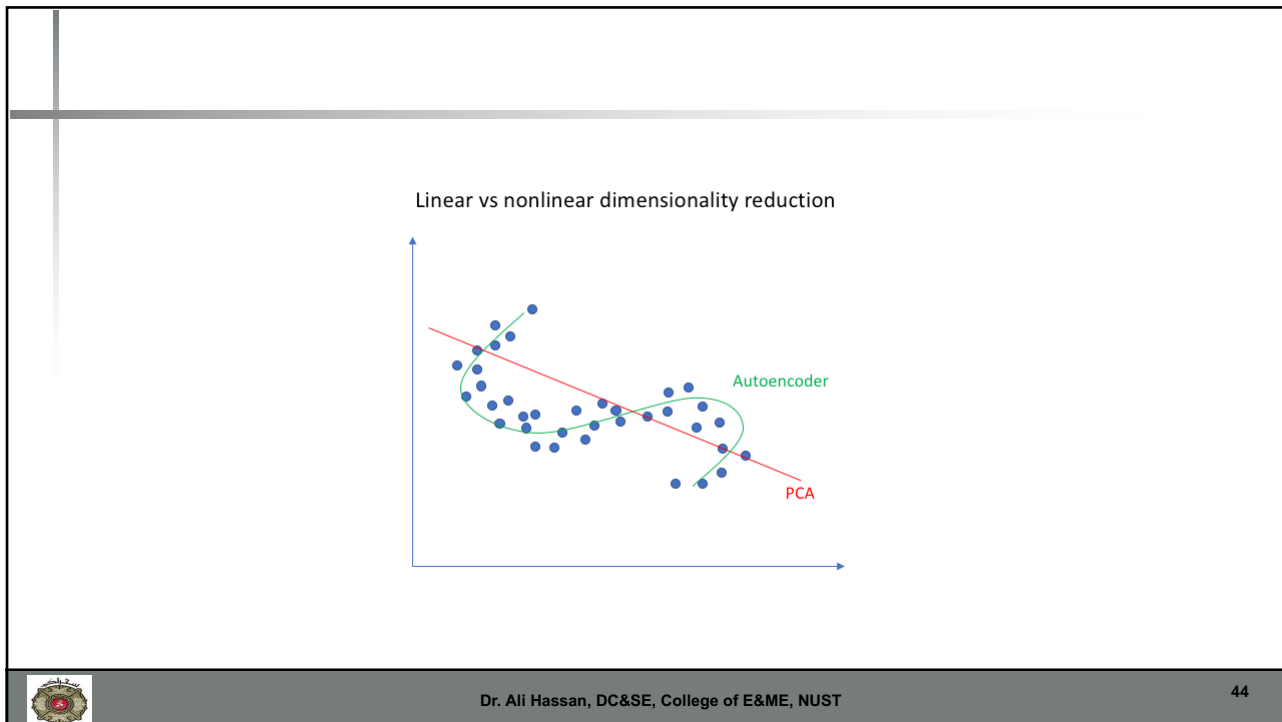
- *In fact, if we were to construct a linear network (ie. without the use of nonlinear activation functions at each layer) we would observe a similar dimensionality reduction as observed in PCA*



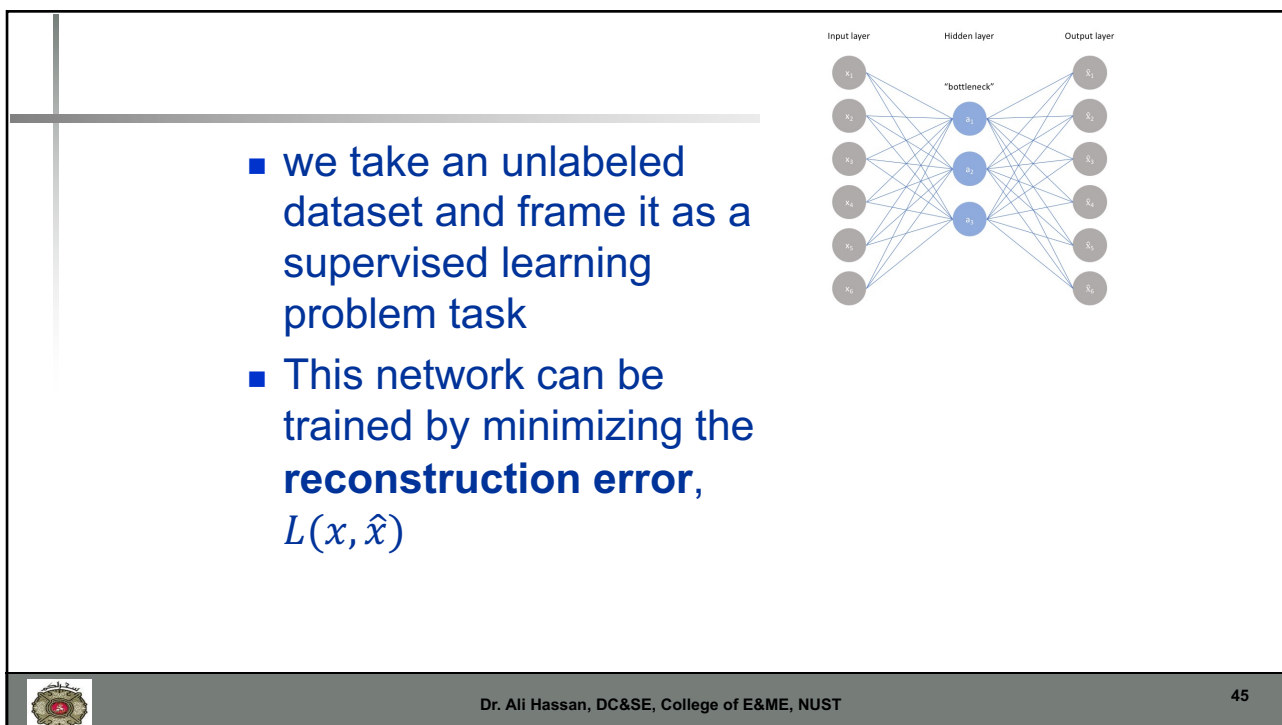
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## The Bottle Neck

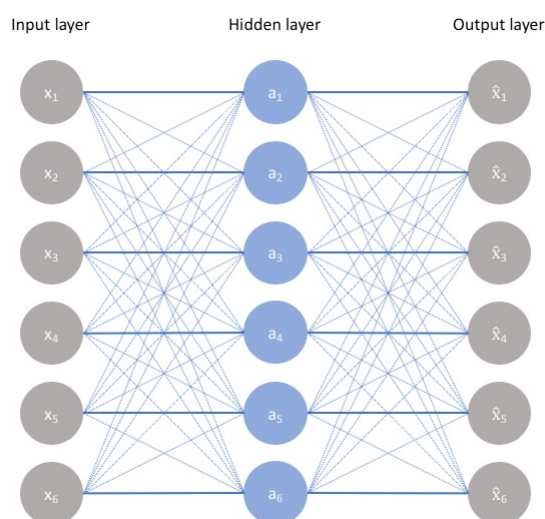
- The **bottleneck** is a key attribute of our network design
- Without the presence of an information bottleneck, our network could easily learn to simply memorize the input values by passing these values along through the network



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## Auto Encoder Tradeoffs

- The ideal autoencoder model balances the following:
  - Sensitive to the inputs enough to accurately build a reconstruction.
  - Insensitive enough to the inputs that the model doesn't simply memorize or overfit the training data.
- $L(x, \hat{x}) + \text{regularizer}$



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## SPARSE AUTOENCODERS



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## Sparse autoencoders

- It uses an alternative method for introducing an information bottleneck without *requiring* a reduction in the number of nodes at our hidden layers.
- We construct our loss function such that we penalize *activations* within a layer



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## Sparse autoencoders

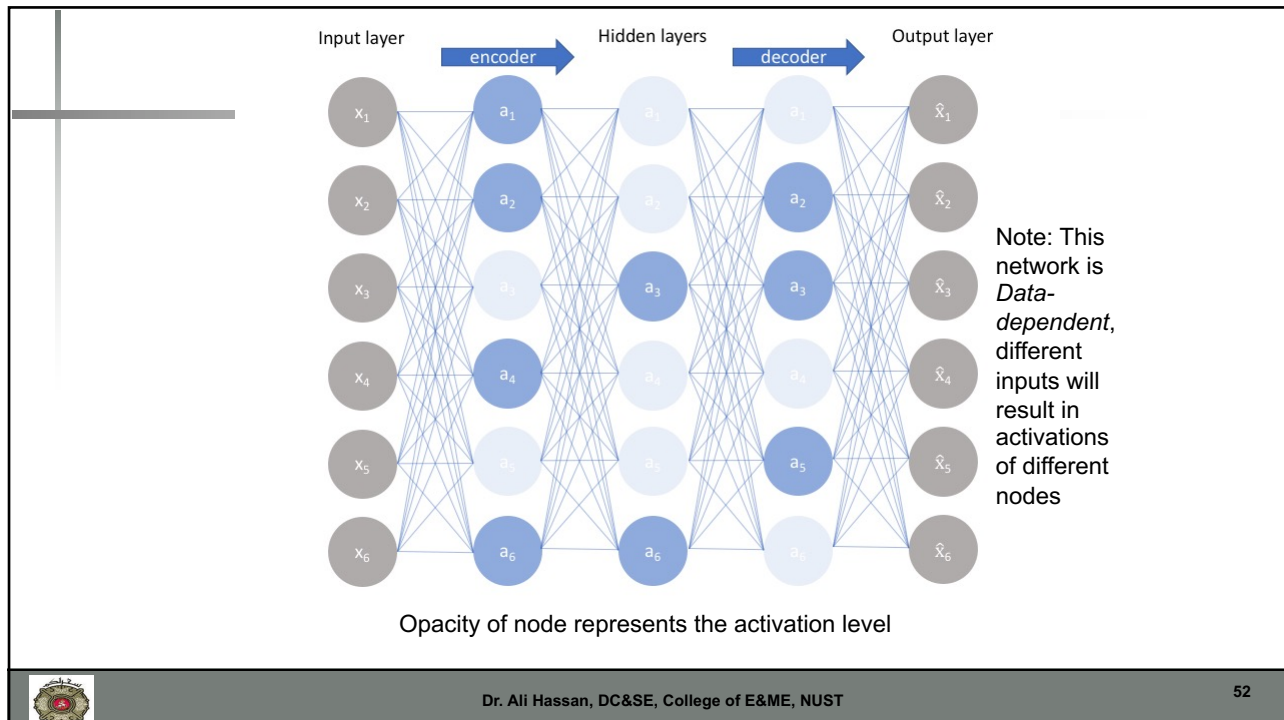
- For any given observation, we'll encourage our network to learn an encoding and decoding which only relies on activating a small number of neurons
- This is a different approach towards regularization, as we normally regularize the *weights* of a network, not the activations.



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## Sparsity Constraint

- Two main ways:
  - **L1 Regularization:**
    - $L(x, \hat{x}) + \lambda \sum_i |a_i^{(h)}|$
  - **KL Divergence**

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## Sparsity Constraint

### ■ KL Divergence

- KL-divergence is a measure of the difference between two probability distributions
- Sparsity parameter  $\rho$  which denotes the average activation of a neuron over a collection of samples
- $\hat{\rho}_j = \frac{1}{m} \sum_i [a_i^{(h)}(x)]$
- by constraining the average activation of a neuron over a collection of samples we're encouraging neurons to only fire for a subset of the observations.
- Now ...
- $L(x, \hat{x}) + \sum_i KL(\rho || \hat{\rho}_j)$



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## DENOISING AUTOENCODERS



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## Denoising autoencoders DAE

- we'd like our autoencoder to be sensitive enough to recreate the original observation but insensitive enough to the training data such that the model learns a generalizable encoding and decoding
- developing a generalizable model is to slightly corrupt the input data but still maintain the uncorrupted data as our target output



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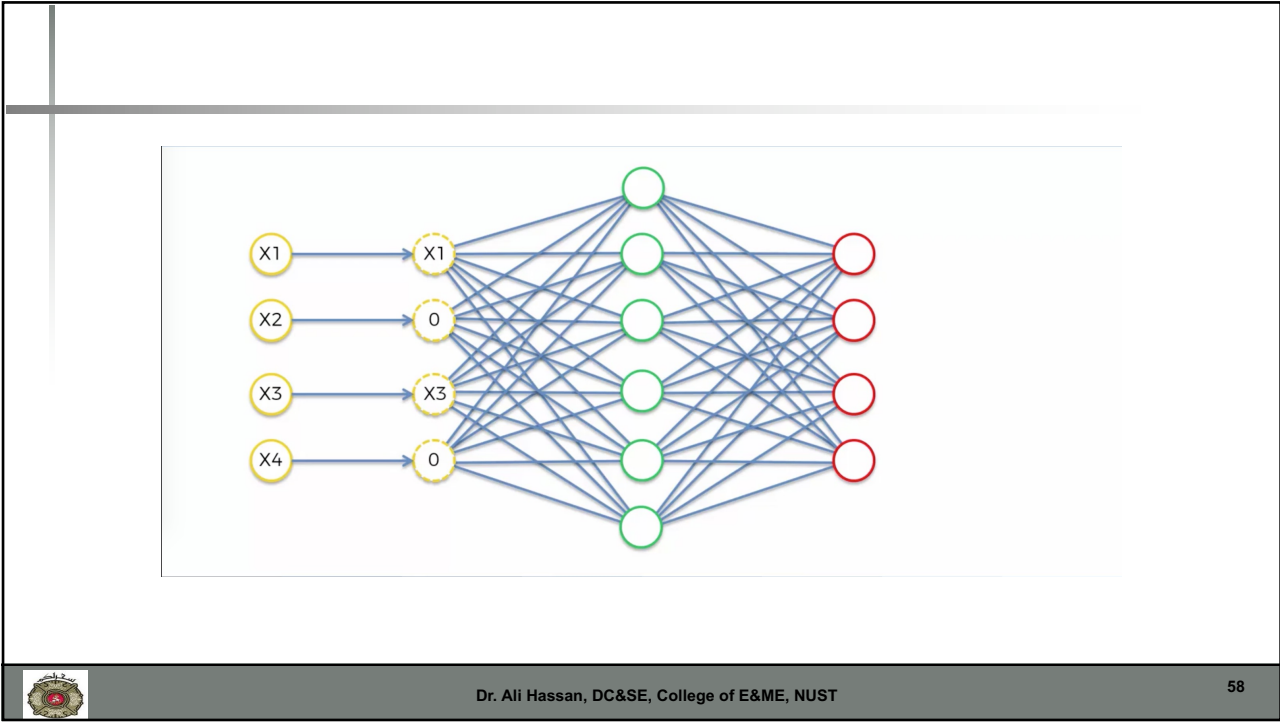
- when there are more nodes in the hidden layer than there are inputs (**over-represented AE**), the Network is risking to learn the so-called “**Identity Function**”, also called “**Null Function**”
- Denoising Autoencoders solve this problem by corrupting the data on purpose by randomly **turning some of the input values to zero**



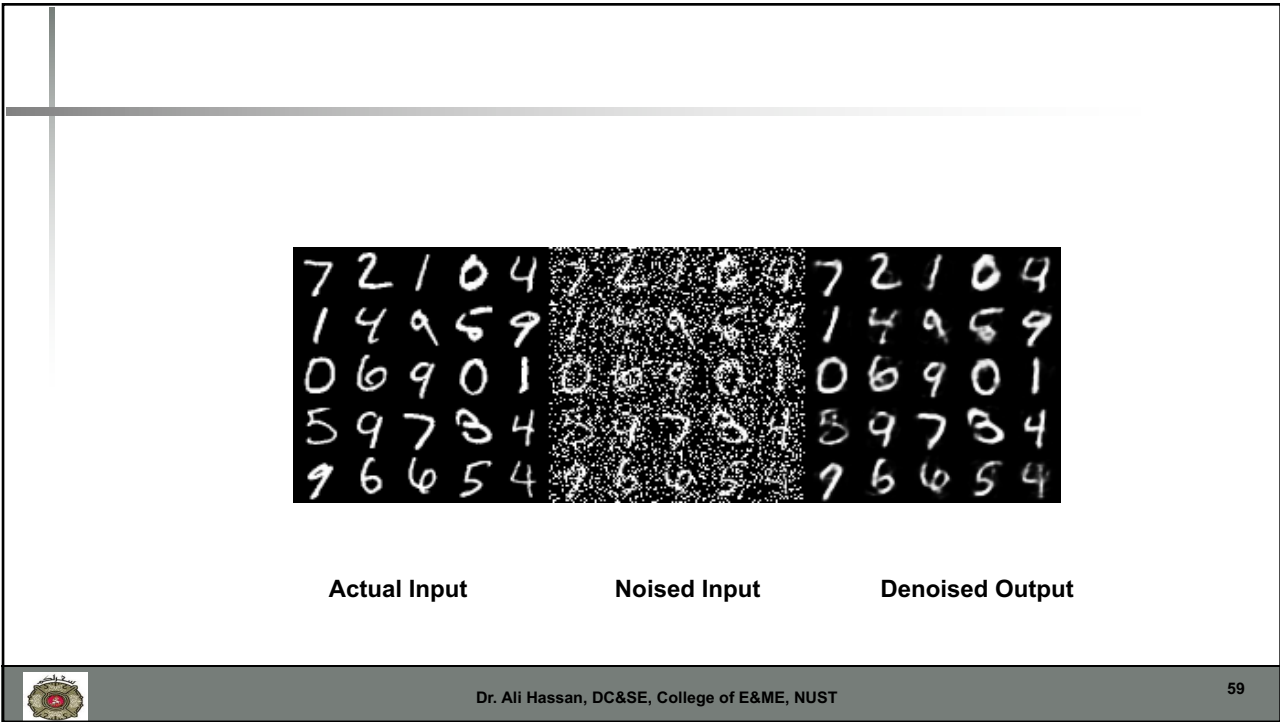
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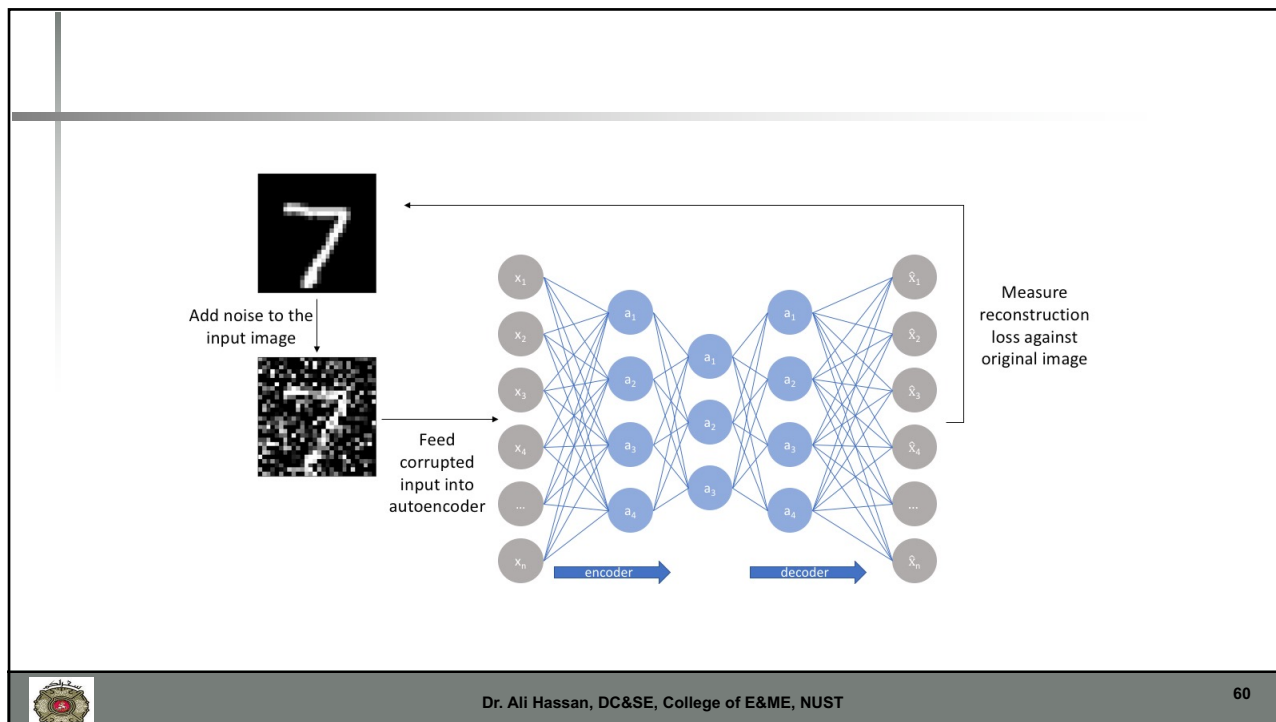


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## DAE

- our model isn't able to simply develop a mapping which memorizes the training data because our input and target output are no longer the same
- the model learns a vector field for mapping the input data towards a lower-dimensional manifold
- if this manifold accurately describes the natural data, we've effectively "canceled out" the added noise

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## Variational Auto Encoders

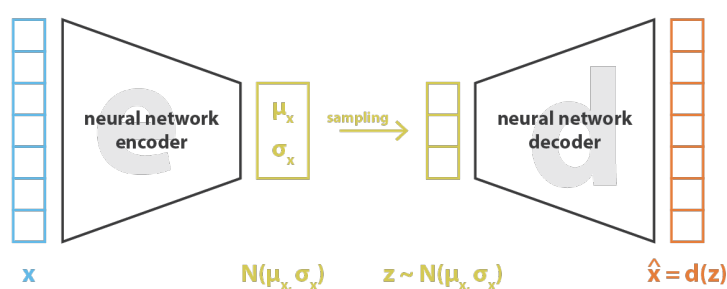
- A VAE can be defined as being an AE whose training is regularised to avoid overfitting and ensure that the latent space enables generative process
- A VAE is composed of an encoder and a decoder, that is trained to minimise the reconstruction error
- instead of encoding an input as a single point, we encode it as a distribution over the latent space



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$$\text{loss} = \|x - \hat{x}\|^2 + \text{KL}[N(\mu_x, \sigma_x), N(0, I)] = \|x - d(z)\|^2 + \text{KL}[N(\mu_x, \sigma_x), N(0, I)]$$



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## VAE Model Training

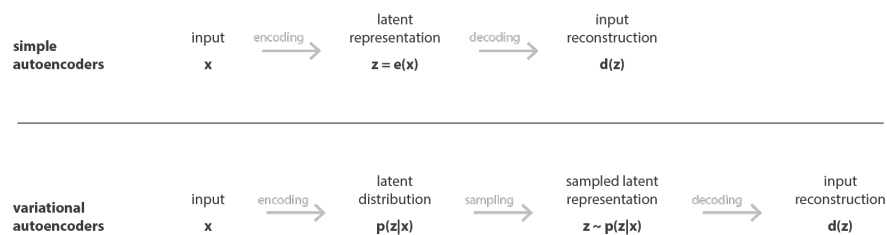
- first, the input is encoded as distribution over the latent space
- second, a point from the latent space is sampled from that distribution
- third, the sampled point is decoded and the reconstruction error can be computed
- finally, the reconstruction error is backpropagated through the network



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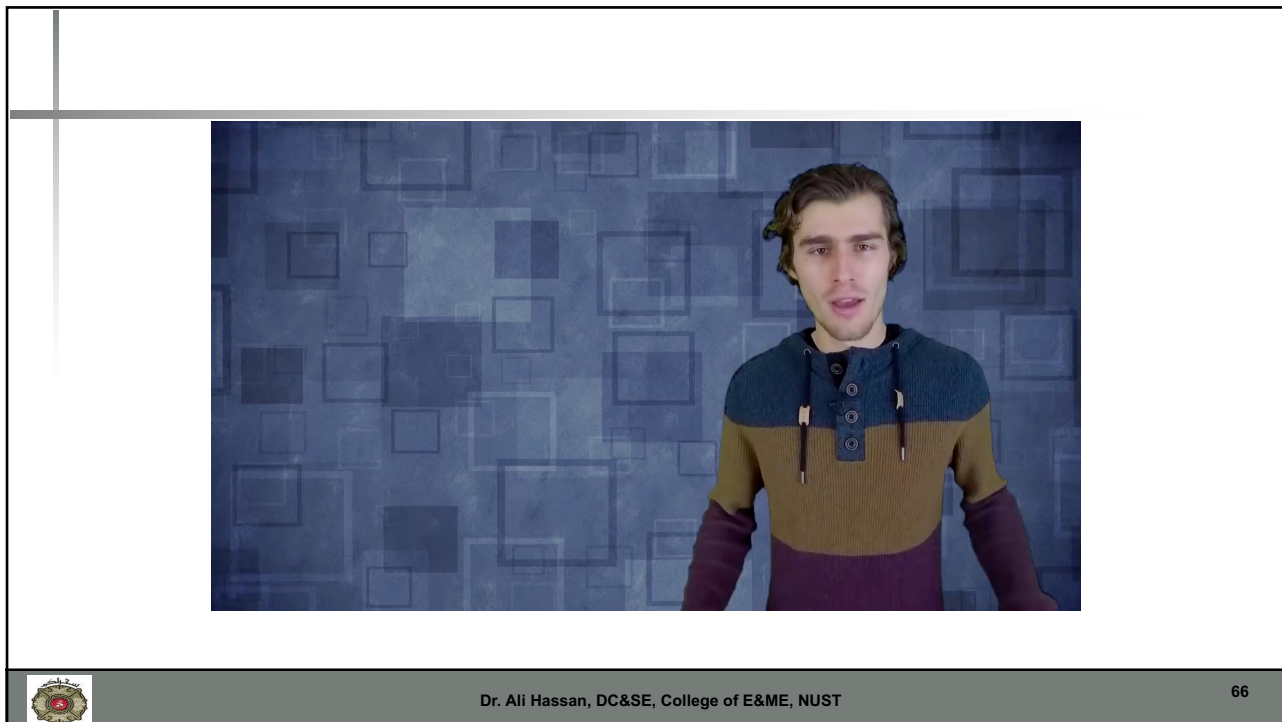
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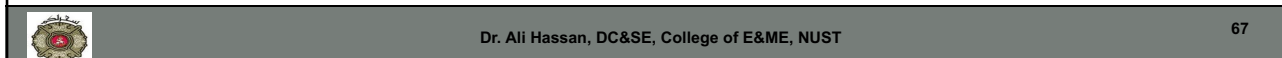
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## Reading Material

- Chapter 14: AutoEncoders Book
  - <https://www.deeplearningbook.org/>
- Tutorial:
  - **Introduction to autoencoders**
  - <https://www.jeremyjordan.me/autoencoders/>



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## Reading Material

- [Chapter 20. Deep Generative Models, Deep Learning, 2016.](#)
- [Chapter 8. Generative Deep Learning, Deep Learning with Python, 2017.](#)
- [Machine Learning: A Probabilistic Perspective, 2012](#)
- NIPS 2016 Tutorial: Generative Adversarial Networks
- <https://www.v7labs.com/blog/autoencoders-guide>



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## Interactive Websites

- <https://www.nvidia.com/en-us/research/ai-demos/>
- <https://thispersondoesnotexist.com/>



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